MILITARY SPECIFICATION

LEAD MONONITRORESORCINATE (LMNR)

1. SCOPE

1.1 This specification covers lead mononitroresorcinate for use in detonators, fuzes, and priming compositions.

2. APPLICABLE DOCUMENTS

2.1 The following documents of this issue in effect on date of invitation, for bids, form a part of this specification to the extent specified herein.

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-109 - Inspection Terms and Definitions.
MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-286 - Propellants, Sampling Inspection and Testing.

PUBLICATIONS

ORDNANCE CORPS

ORD-M608-11 - Procedures and Tables for Continuous Sampling by Attributes.

(Copies of specifications, standards, drawings and publications required by contractors in connection with specific procurement functions, should be obtained from the procuring activity or as directed by the contracting officer.)

FSC: 1375

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2.2 Other publications.-The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

Interstate Commerce Commission
49 CFR 71-78 - Interstate Commerce Commission Rules and Regulations for the Transportation of Explosives and Other Dangerous Articles.

(The Interstate Commerce Commission Regulations are now a part of the Code of Federal Regulations (1949 edition-revised 1950) available from the Superintendent of Documents, Government Printing Office, Washington 25, D. C. Orders for the above publication should cite "49 CFR 71-78 (Current Revision)").

3. REQUIREMENTS

3.1 Color and appearance.-The lead mononitroresorcinate shall be light brown in color when tested as specified in paragraph 4.3.1.

3.2 Chemical and physical properties.-Lead mononitroresorcinate shall conform to the limits for the properties specified in Table I, when tested as specified in the applicable test paragraph.

Table I

Properties	Minimum (min.)	Maximum (max.)	Test Paragraphs
Apparent Density, grams (gm.) per milliliter (ml.)	0.20	0.35	4.3.2
Acidity, pH	5.0	6.5	4.3.3
Lead Content, percent	70.5	72.5	4.3.4
Nitro group content, percent	7.2	7.6	4.3.5
Vacuum Stability 120 degree Centigrade (°C.), (ml.) (in 40 hours)		0.4	4.3.6

3.3 Workmanship.-The lead mononitroresorcinate shall be free from grease, wood slivers or chips, rubber or plastic particles or any other foreign material or particles.

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4. QUALITY ASSURANCE PROVISIONS

4.1 General quality assurance provisions.-The supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examinations and tests shall be kept complete and available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Reference shall be made to Standard MIL-STD-109 in order to define the terms used herein. Inspection shall be performed in accordance with this specification and other specifications referenced in any of the contractual documents.

4.1.1 Contractor quality assurance system.-If the contractor desires to utilize a quality assurance system, which is at variance with the quality assurance provisions of 4.2 and 4.3 and other documents referred to herein, he shall submit a written description of the system to the contracting officer for approval prior to initiation of production. It shall include a description covering controls for lot formation and identification, inspections to be performed, inspection stations, sampling procedures, methods of inspection, (measuring and testing equipment), and provisions for control and disposition of non-conforming material. The written description will be considered acceptable when, as a minimum, it provides the quality assurance provisions required by the provisions of 4.2 and 4.3 and the other documents referred to herein. The contractor shall not be restricted to the inspection station or the method of inspection listed in this specification provided that an equivalent control is included in the approved quality assurance procedure. In cases of dispute as to whether certain procedures of the contractor's system provide equal assurance, the comparable procedure of this specification shall apply. The contractor shall notify the Government of, and obtain approval for any changes to the written procedure that effects the degree of assurance required by this specification or other documents referred to herein.

4.1.2 Submission of product.-At the time the completed lot of product is submitted to the Government for acceptance, the contractor shall supply the following information accompanied by a certificate which attests that the information provided is correct and applicable to the product submitted:

a. A statement that the lot complies with all quality assurance provisions of the approved current written description of the system.

b. Quantity of product inspected.

c. Results obtained for all inspection performed.

d. Specification number and date, together with an identification and date of changes.

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e. Certificates of analysis on all material referred to by Government specifications, procured directly by the contractor.

f. Quantity of product in the lot.

g. Date submitted.

The certificate shall be signed by a responsible agent of the certifying organization. The initial certificate submitted shall be substantiated by evidence of the agent's authority to bind his principal. Substantiation of the agent's authority will not be required with subsequent certificates unless, during the course of the contract, this authority is vested in another agent of the certifying organization.

4.1.3 Government verification.-Using the contractor's written quality assurance procedure (see 4.1.1), this detailed specification, and other contractual documents as a guide, the Government inspector shall verify all quality assurance operations performed by the contractor. Verification shall be in accordance with a or b as applicable, the decision being the responsibility of the procuring activity. In either case, the inspector shall also ascertain, prior to acceptance, that all quality assurance provisions of other specifications referred to in any of the contractual documents have been complied with. Deviations from prescribed or agreed upon procedures discovered by the Government inspector shall be brought to the attention of the supplier. Disposition of the product and remedial action shall be as directed by the Government inspector and, depending on the nature of the deviation, may consist of lot rejection, screening, re-sampling, re-instruction of the supplier's employees, or other appropriate action:

a. Verification at the point of manufacture shall be accomplished at unscheduled intervals in accordance with 4.1.3.1 and 4.1.3.2.

b. Verification at the point of delivery shall be in accordance with 4.1.3.2.

4.1.3.1 Surveillance.-Surveillance shall include, but is not limited to:

a. Observation of procedures concerning the lot formation and identification.

b. Observation of sampling procedures and application of acceptance criteria.

c. Determination that all required examinations and tests are performed in accordance with the prescribed procedures of this specification, or approved equivalents thereto.

d. Review of procedures for control and disposition of non-conforming material.

4.1.3.2 Product inspection.-Product inspection shall consist of Government inspection of product which has been previously inspected by the contractor and found to meet the quality assurance provisions of this specification. The inspection by the Government shall be performed in order to determine that the product is of the quality required by this specification and that the contractor's records are reliable.

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4.2 Inspection provisions.-

4.2.1 Lot formation.-A lot shall consist of one or more batches of lead mononitroresorcinate produced by one manufacturer, in accordance with the same specification, or same specification revision, under one set of operating conditions. Each batch shall consist of that quantity of lead mononitroresorcinate that has been subjected to the same unit chemical or physical process intended to make the final product homogeneous.

4.2.2 Examination.-Sampling plans and procedures for the following classifications of defects shall be in accordance with Standard MIL-STD-105 except that inspection for critical defects shall be 100 percent. Continuous sampling plans in accordance with Handbook ORD-M608-11 may be used if approved by the procuring activity. Also, at the option of the procuring activity, AQL's and sampling plans may be applied to the individual characteristics listed using an AQL of 0.25 percent for each major defect and an AQL of 0.40 percent for each major defect and an AQL of 0.40 percent for each minor defect.

4.2.2.1 Bag, cotton duck (see 5.1.1).-

Categories	Defects	Method of inspection
Critical:	None defined.	
Major:	AQL 0.40 percent	
101.	Lead mononitroresorcinate improperly or insufficiently wetted	Visual
102.	Bag pierced or torn	Visual
103.	Cloth cap pierced, torn or improperly positioned	Visual
104.	Bag improperly closed	Visual
Minor:	None defined.	

4.2.2.2 Bag, waterproof (see 5.1.1).-

Categories	Defects	Method of inspection
Critical:	None defined.	
Major:	AQL 0.40 percent	
101.	Bag pierced or torn	Visual
102.	Cotton duck bag missing	Visual
103.	Bag improperly closed	Visual
Minor:	None defined.	

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4.2.2.3 Container prior to closing (see 5.2.1).-

Categories	Defects	Method of inspection
Critical:	None defined.	
Major:	AQL 0.65 percent	
101.	Jute liner torn	Visual
102.	Sawdust insufficiently or improperly saturated	Visual
103.	Filled waterproof bag incompletely or improperly surrounded by sawdust	Visual
104.	Jute liner improperly closed (sawdust escaping)	Visual
105.	Container pierced, cut or damaged	Visual
Minor:	None defined.	

4.2.2.4 Closed container (see 5.2.1).-

Categories	Defects	Method of inspection
Critical:	None defined.	
Major:	AQL 0.25 percent	
101.	Container improperly closed or sealed	Visual-Manual
102.	Marking misleading or unidentifiable	Visual
Minor:	AQL 1.00 percent	
201.	DOD symbol misleading or unidentifiable	Visual
202.	Bare spot on exterior other than slight scratches (applicable metal containers)	Visual

4.2.3 Testing.-

4.2.3.1 Determination of physical and chemical properties (see 3.1 through 3.3).

4.2.3.1.1 Sampling by the lot.-A primary sample shall be taken from each barrel or drum in the lot.

4.2.3.1.2 Preparation of primary samples.-Equal portions shall be removed by means of a horn spoon, from each bag in the barrel or drum to form a primary

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sample of approximately 100 gm. (dry weight). The sample shall be blended on a smooth surface by mixing with a horn spatula. Water shall be added in case the sample appears dry. The sample shall be spread out and divided into squares approximately 1/2 inch on a side, by means of the spatula, small portions shall be taken from each square to make a retained sample of approximately 20 gms. (dry weight). These portions, thoroughly wetted, shall be transferred to a smooth-necked bottle, tightly stoppered with a rubber stopper, and the stopper secured with adhesive tape. This retained sample bottle shall be so labeled that the barrel or drum from which the material was taken can be easily identified. These retained samples of each primary sample shall be held for possible future examination. The remaining portions of all the primary samples shall be used to form a composite sample.

4.2.3.1.3 Preparation of composite sample.-The remaining primary portions shall be combined, thoroughly mixed spread out, marked into squares and small equal portions taken from each square until a composite sample of approximately 50 gm. (dry weight) is obtained. This sample, thoroughly wetted, shall be transferred to a smooth-necked bottle tightly stoppered with a rubber stopper and the stopper secured with adhesive tape. The bottle shall be labeled to show the name of material, manufacturer, plant, contract or order number, lot number and number of pounds in the lot. The tests specified in 4.3 shall be performed on this composite sample. If the sample fails to comply with the requirements specified herein, the lot shall be rejected.

4.3 Test methods and procedures.-The following tests shall be performed:

4.3.1 Color and appearance.-

4.3.1.1 Microscopic procedure.-Transfer a portion of about 10 gm. of the wet sample as received to a Buchner funnel and apply suction until the sample is almost dry. Spread the sample on a watch glass and place in an oven at 70-75°C. for at least four hours or until constant weight is obtained. Take four portions of about 0.5 milligrams (mg.) each from different sections of the sample and spread them over an area of a 2 centimeters (cm.) square on a glass microscopic slide. Examine the material using a magnification of about 30 times.

4.3.2 Apparent density.-Into a 50 ml. beaker, weigh 2.00 ± 0.01 gm. of dry LMNR. Carefully add 95 percent ethyl alcohol until a thick paste is formed on mixing with a plastic rod. Using the plastic rod and a stream of 95 percent ethyl alcohol transfer the paste to a 25 ml. graduate having an inside diameter of 20 millimeter. Cover the graduate with the thumb and shake thoroughly until the sample is completely wetted. After shaking the graduate, allow it to stand for two hours away from vibrations and jarring. Read the settled volume and calculate the apparent density using the following equation:

$$\text{Apparent Density} = \frac{W}{V}$$

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where:

W = weight of the sample in gms.
V = observed volume of sample in mls.

4.3.3 Acidity.-Transfer 1.00 gm. of dry LMNR to a 150 ml. beaker. Add 100 ml. of freshly boiled and cooled distilled water having a pH of 6.0 ± 0.5 and allow the mixture to stand, with occasional stirring, for 15 minutes. Filter and determine the pH of the filtrate at 25° ± 2°C. by means of a pH meter which reads directly in pH units and is capable of measuring accurately within 0.1 pH unit. The pH meter shall be equipped with a glass electrode and a saturated calomel electrode. It shall be calibrated with standard buffer solutions.

4.3.4 Lead content.-A sample of 0.5 gm. shall be weighed to the nearest .0001 gm. and placed in a 300 ml. tall form beaker. The sample shall be wet with 5 ml. of distilled water. Then 20 ml. of dilute sulfuric acid and 5 ml. of concentrated nitric acid shall be added. The beaker, with a ribbed watch glass as a cover, shall be placed on a heater. The heater shall be turned on low heat. The solution shall be evaporated until the liquid is clear and dense white fumes are given off. The heat shall be increased gradually during the evaporation in order to keep the solution fuming. After evaporating to the desired state, the beaker and contents shall be cooled and 10 ml. of concentrated nitric acid shall be added. The solution shall again be evaporated until the liquid is clear and dense white fumes are given off. After the second evaporation the beaker and contents shall be cooled, the watch glass and side of the beaker shall be washed down with distilled water. The contents shall be made up to 100 ml. and allow to stand for 2 hours. The precipitate shall be filtered using a tared Gooch crucible. The precipitate shall be transferred to the crucible using a 50 percent solution of alcohol. The precipitate shall be washed three times with a 50 percent alcohol solution. The crucible and contents shall be dried for one hour at 100°C., cooled in a desiccator and weighed. The percent lead shall be calculated as follows:

$$\text{Percent lead} = \frac{68.32A}{B}$$

where:

A = weight of the precipitate
B = weight of the sample

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4.3.5 Nitro-group content.

4.3.5.1 Titanous chloride method.

4.3.5.1.1 Procedure.-Transfer approximately 0.2 gm. of dry LMNR, weighed within 0.1 mg., into a titration flask. Add 25 ml. of 95 percent ethyl alcohol and a few glass beads to the flask and allow carbon dioxide to pass slowly through the flask for about ten minutes in order to displace the air in the flask. Continue the flow of carbon dioxide through out the determination. Then add 50 ml. of 0.2 normal (N) titanous chloride solution from the burette. Connect the flask to a water cooled condenser, place it on a hot plate and boil gently for about five minutes. Increase the flow of carbon dioxide and place the flask in a water bath to cool. Disconnect the condenser and titrate the solution with 0.15N ferric ammonium sulfate solution until the end point, as shown by the disappearance of the dark color of the titanium. Add 10 ml. of ammonium thiocyanate indicator solution and continue the titration with ferric ammonium sulfate solution until a reddish color, which remains for one minute, is obtained.

Conduct a blank determination following the above procedure with exactly the same quantity of reagents and exactly the same quantity of 0.2N titanous chloride solution.

Calculate the percentage of nitro-group in the LMNR as follows:

$$\text{Percent nitro-group} = \frac{0.7668\,(A-B)N}{W}$$

where:

A = volume of ferric ammonium sulfate solution used for blank, ml.
B = volume of ferric ammonium sulfate solution used for LMNR
N = normality of ferric ammonium sulfate solution
W = weight of the LMNR sample, gm.

4.3.5.2 Alternate method, titanous sulfate method.

4.3.5.2.1 Titanous sulfate solution.-Two hundred ml. of 20 percent titanous sulfate solution shall be filtered through asbestos or filter paper. Then add 160 ml. of 1:1 sulfuric acid and cool under an atmosphere of carbon dioxide. The volume shall be made up to one liter with cool distilled water (which has been boiled for 20 to 30 minutes and cooled under an atmosphere of carbon dioxide). Pass a rapid stream of carbon dioxide through the solution for an hour to thoroughly mix the solution. Allow the solution to age for several days under pressure of carbon dioxide before using.

4.3.5.2.2 Standardization of solution.-The titanous sulfate shall be standardized against paranitraniline. Weigh to the nearest 0.0001 gm. about 0.2 gm. of paranitraniline, (with a melting point of 146.5°C. or higher) previously dried at 100 C. for one hour, into a titration flask. Add 25 ml. of 95 percent ethyl

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alcohol to the flask and allow carbon dioxide to pass slowly through the flask for about 10 minutes in order to displace the air in the flask. Continue the flow of carbon dioxide throughout the determination. Place a few glass beads in the flask, connect the flask to the condenser, and pass in a rapid stream of carbon dioxide. Boil gently until refluxing commences, reduce the flow of carbon dioxide to about 2 bubbles per second, and continue boiling for 5 minutes. Remove flask from heater, increase the gas stream, and cool the flask rapidly under the tap. Disconnect the flask from the condenser and run in 0.15N ferric ammonium sulfate solution from a burette until the dark color of the titanium disappears. Add 10 ml. of ammonium thiocyanate solution as an indicator and continue the titration with ferric ammonium sulfate until a reddish color is obtained, which remains permanent for one minute. Apply burette and temperature corrections.

4.3.5.2.3 Comparison of titanous sulfate and ferric ammonium sulfate solution.-Place 25 ml. of 95 percent ethyl alcohol in a titration flask and pass carbon dioxide through for 10 minutes to displace the air. Add 15 to 20 ml. of titanous sulfate and proceed as described in 4.3.5.2.2.

Calculate the ratio R as follows:

$$R = \frac{A}{B}$$

where:

R = ratio of titanous sulfate to ferric ammonium sulfate.
A = ml. of titanous sulfate used.
B = ml. of ferric ammonium sulfate used.

4.3.5.2.4 Procedure.-The determination of nitro group in the specimen of lead mononitroresorcinate shall be performed exactly as described in 4.3.5.2.2 using a 0.2 gm. specimen in place of the paranitraniline. The percent of nitro group shall be calculated as follows:

$$\text{Percent nitro group} = \frac{33.33\ P\ (D-RC)}{W\ (E-RF)}$$

where:

P = weight of paranitraniline used in 4.3.5.2.2.
D = ml. of titanous sulfate used in 4.3.5.2.4.
R = Ratio determined in 4.3.5.2.3.
C = ml. of ferric ammonium sulfate used in 4.3.5.2.4.
W = weight of specimen used.
E = ml. of titanous sulfate used in 4.3.5.2.2.
F = ml. of ferric ammonium sulfate used in 4.3.5.2.2.

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4.3.6 120°C. vacuum stability.-The vacuum stability test shall be in accordance with Standard MIL-STD-286 Method 403.1, except the weight of LMNR shall be 2.3 gm. and a temperature of 120°C. shall be used.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging.

5.1.1 Level A.-Lead mononitroresorcinate shall be wet with not less than 20 percent of a solution of denatured ethyl alcohol and water containing not less than 50 percent of ethyl alcohol, and in this wet condition shall be placed in bags of 4 ounce (or heavier) duck, each bag containing approximately 25 pounds (dry weight) of lead mononitroresorcinate. There shall be placed in each bag, and over the lead mononitroresorcinate, a cap of the same cloth and diameter as the bag shall be securely tied.

5.1.2 Level B.-Unless otherwise specified, lead mononitroresorcinate shall be packaged as specified in 5.1.1.

5.1.3 Level C.-Unless otherwise specified, lead mononitroresorcinate shall be packaged as specified in 5.1.1.

5.2 Packing.

5.2.1 Level A.-Five of the bags described in 5.1 shall be placed in a strong waterproof bag, and this bag securely tied. The large waterproof bag shall contain not more than 100 pounds (dry weight) of lead mononitroresorcinate. This bag shall be placed in the center of a container complying with specifications, 5 and 5B (metal barrels or drums), 17H (metal drums, singel trip) or 10B (wooden barrels or kegs), of Regulations for Transportation of Explosive and other Dangerous Articles, etc. Each container shall be lined with a heavy, close fitting bag made of jute, or other suitable bag material of equal strength. The large waterproof bag shall be completely surrounded, within the jute bag by not less than 3 inches of well-packed sawdust saturated with the same 50 percent solution of ethyl alcohol, and the jute bag shall be closed by securely sewing to prevent escape of sawdust. The outer container (drum or barrel) shall be watertight.

5.2.2 Level B.-Unless otherwise specified, lead mononitroresorcinate shall be packed as specified in 5.2.1.

5.2.3 Level C.-Unless otherwise specified, lead mononitroresorcinate shall be packed as specified in 5.2.1.

5.3 Marking.-In addition to any special marking required by the contract or Purchase Order, containers shall be marked in accordance with Interstate Commerce Commission Rules and Regulations, etc. (Code of Federal Regulation 49 CFR 71-78) and MIL-STD-129. Each barrel, box, drum or keg shall be plainly marked "INITIATING EXPLOSIVE - DANGEROUS - DO NOT STORE OR LOAD WITH ANY HIGH EXPLOSIVES."

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6. NOTES

6.1 Ordering data.-Procurement documents should specify the title, number and date of this specification.

6.2 Intended uses.-This specification covers lead mononitroresorcinate intended for uses an an initiator in detonators, fuzes and priming compositions.

Notice.-When Government drawings specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any was supplied said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

Custodian:
Army-Ordnance Corps

Preparing activity:
Army-Ordnance Corps

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